

# Vikas Verma

Software Engineer | Node.js & Microservices | Event-Driven Architectures

Noida, India | +91 9161083119 | [vikasvermasde@gmail.com](mailto:vikasvermasde@gmail.com) | [LinkedIn](#)

## SUMMARY

---

I reduce latency. At Rannkly, I cut API response times by 40% and database query latency by 50% — on live, high-traffic SaaS systems. I build backend infrastructure that doesn't break when it matters most.

For around 7 years, I've architected systems that handle real-scale, queue-based workflows with Redis & BullMQ, fault-tolerant microservices, payment integrations across Stripe, Paystack, and PayPal, and AI-powered automation that replaced 80% of manual processing effort.

What I actually do well:

- Design async, event-driven backends that stay reliable under pressure
- Optimize databases (MongoDB + MySQL) before they become the bottleneck
- Own systems end-to-end — from TRDs and architecture decisions to production firefighting
- Ship clean, tested Node.js, PHP code (SOLID, code reviews) that teams don't curse at later

## TECHNICAL SKILLS

---

**Languages:** JavaScript, TypeScript, PHP

**Backend Technologies:** Node.js, Express.js, Laravel, REST APIs, Microservices, MVC Architecture

**Databases:** MySQL, MongoDB, SQL, NoSQL, RDBMS

**Caching & Messaging:** Redis, BullMQ, Caching, Queue-Based Processing

**Cloud & AWS Services:** AWS, AWS Lambda, S3, API Gateway

**Architecture & Design:** Event-Driven Architecture, Concurrency Control, Retry Mechanisms, OOP, SOLID Principles

**ORM & Libraries:** Sequelize, Mongoose, JWT

**Testing:** Jest, Unit Testing

**DevOps & Tools:** Docker, Git, GitHub, Jira

**Development Practices:** Agile Methodologies, Problem-Solving, System Design

**AI-Assisted Development:** ChatGPT, Claude, Gemini, Cursor IDE, Antigravity IDE

## EXPERIENCE

---

**RANNKLY — Senior Software Engineer** (07/2024 – 04/2026)

*Tech Stack: Node.js, Express.js, MongoDB, MySQL, Redis, BullMQ, Docker, Jira*

- Architected asynchronous job processing pipelines using Redis and BullMQ, incorporating retry logic and idempotency to achieve fault-tolerant, concurrent task execution at scale.
- Developed scalable RESTful APIs and microservices with Node.js and Express.js, applying event-driven architecture to enable decoupled, high-throughput backend systems.
- Streamlined system performance through Redis caching, rate limiting, and SQL/NoSQL query tuning, reducing API latency and improving capacity for high-concurrency workloads.
- Engineered a real-time notification system, resolving critical throughput bottlenecks and achieving a 10x improvement in system capacity and reliability.
- Authored TRDs and POCs, led backend architecture decisions, and integrated AI-assisted workflows (ChatGPT, Claude, Gemini) to accelerate delivery and boost individual productivity by ~30%.
- Leveraged AWS S3 for scalable file storage and AWS Lambda for serverless function execution, integrating cloud infrastructure into core backend workflows to improve scalability and reduce server overhead.

**WeAssemble — Backend Engineer** (04/2022 – 07/2024)

*Tech Stack: Node.js, ExpressJS, MySQL, MongoDB, BullMQ, Docker, Jira*

- Architected and maintained RESTful APIs using [Node.js](#), [Express.js](#), and Laravel, supporting critical business workflows and data-intensive operations across interconnected backend systems.
- Structured normalized relational schemas and enhanced MySQL queries and MongoDB aggregation pipelines, reducing database latency by 35% through indexing and query restructuring.
- Enforced JWT-based authentication and role-based access control (RBAC), ensuring secure and compliant access management across backend services.
- Collaborated cross-functionally to define API contracts, contributed to architecture discussions, code reviews, and Agile ceremonies to maintain delivery quality and team alignment.

### **BabySoft Software — Backend Engineer Intern (02/2019 – 10/2021)**

*Tech Stack: PHP, Laravel, [Node.js](#), [Express.js](#), MySQL, MongoDB*

- Built and delivered 100+ RESTful APIs using PHP and Laravel to power dynamic web applications and backend business logic across multiple client projects.
- Created and improved relational and NoSQL database schemas in MySQL and MongoDB, enforcing data integrity and enhancing query performance in production environments.
- Translated client requirements into high-fidelity UI mockups and prototypes, reducing revision cycles and aligning product direction with stakeholder expectations.
- Partnered with cross-functional teams across design, development, and stakeholder groups to drive requirement gathering, solution design, and project presentations.
- Executed end-to-end QA, including functional, performance, and compatibility testing; managed deployments via Git to ensure stable, regression-free production releases.
- Integrated third-party APIs and external services into Laravel-based backends, resolving performance bottlenecks and improving response times across multiple client-facing applications.

## **EDUCATION**

---

### **Bachelor of Technology in Computer Science and Engineering**

06/2018 - 05/2021

Institute of Technology and Management - Lucknow, India

### **Diploma in Computer Science and Engineering**

06/2014 - 05/2017

Government Polytechnic Saharanpur - Saharanpur, India

## **PROJECTS**

---

### **RANNKLY — Reputation and Report Management Platform**

, Noida, India

*Tech Stack: Node.js, Typescript, React, Redis, BullMQ, MySQL, MongoDB, Jest, Docker, AWS*

Engineered scalable backend services and RESTful APIs for a high-volume reputation management platform, handling thousands of data processing workflows daily.

- Architected a reporting system capable of generating thousands of reports daily with fault-tolerant delivery pipelines.
- Cut database response time by 50% through query tuning, indexing strategies, and streamlined data retrieval.
- Built automation and monitoring pipelines for report workflows, ensuring operational stability at scale.
- Strengthened deployment reliability through unit and integration testing with Jest.

### **EVA — AI-Powered Booking Automation System**

, Mumbai, India

*Tech Stack: Node.js, Nest.js, Typescript, Email Automation, ExpressJS, MySQL, MongoDB, Docker, AWS*

Created a backend system automating manual booking and communication for high-volume operations.

- Increased operational efficiency by 80% by replacing manual workflows with AI-powered automation.

- Improved customer service productivity by 90% through streamlined email booking and response handling.
- Built intelligent email parsing and automated response workflows, reducing booking turnaround time significantly.
- Scaled backend workflow systems with cross-functional teams to handle high-volume operational loads.

### **WANZY — Payment Microservice**

, **Mumbai, India**

**Tech Stack:** *Node.js, Nest.js, Typescript, Webhooks, ExpressJS, MySQL, MongoDB, Docker, Stripe · Paystack · PayPal*

Developed a Payment Service in a microservices ecosystem for secure, scalable transactions with multiple providers.

- Integrated Stripe, Paystack, and PayPal with webhook handling and retry mechanisms, increasing payment success rates by 30%.
- Established asynchronous processing patterns and service-to-service communication for reliability across distributed systems.
- Integrated idempotent transaction workflows to prevent duplicate charges and ensure data consistency.

### **ROAMY — Event & Hotel Booking Platform**

, **Mumbai, India**

**Tech Stack:** *[Node.js](#), [Nest.js](#), JavaScript, MySQL, MongoDB, REST APIs, Docker, AWS, Stripe*

Built and improved backend modules for a full-featured booking platform supporting search, comparison, and reservation management for events and hotels.

- Contributed to a 30% increase in customer satisfaction through refined booking workflows and backend performance improvements.
- Created efficient service integrations and intuitive API flows to strengthen platform usability.
- Partnered with product and engineering teams to deliver scalable, high-performance booking solutions.

### **Villa Manager — Hotel Booking Platform**

, **Lucknow, India**

**Tech Stack:** *PHP, Laravel, Redis, MySQL, JQuery, JavaScript*

Delivered the backend of a hotel booking platform, building REST APIs, booking workflows, authentication, payment integrations, and scalable services while optimizing performance and reliability.

- Built and maintained the backend architecture for a hotel and event booking platform.
- Planned and executed RESTful APIs for search, booking, user management, and payment workflows.
- Streamlined booking and search workflows to improve performance and user experience.
- Implemented authentication, authorization, payment gateway integrations, and third-party API integrations.
- Collaborated with frontend, QA, and product teams to deliver new features and resolve production issues.

### **Skill4Study — A Learning Platform**

, **Lucknow, India**

**Tech Stack:** *PHP, Laravel, MySQL, JQuery, JavaScript*

- Produced the backend for an English learning platform designed to improve Reading, Writing, Listening, and speaking skills through interactive exercises.
- Built RESTful APIs for user management, progress tracking, assessments, and personalized learning workflows.
- Introduced exercise modules, scoring systems, and performance analytics to track learner progress.
- Designed scalable backend services and optimized database queries to ensure reliable performance and a seamless learning experience.